Control work 6, and work on the mistakes

The logarithmic block in one paper — where the domain line is worth more than the arithmetic.

LESSON 69–70 | 2 × 40 MIN | ALGEBRA 10, NAZORAT ISHI 6 | P2 · CHAPTER 2 REVIEW

LESSON CLOCK

3 | 40' | 10' | 25' | 12'

Instructions	3 min
The paper	40 min
Self-mark	10 min
Rewrite	25 min
The domain drill	12 min

LEARNING OBJECTIVES

- Apply the logarithmic methods under time.
- State a domain and intersect with it without prompting.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 297–300
Cambridge	Pure Mathematics 2 & 3, pp. 53–54
Workbook	Control paper F

01 Explanation

The paper — 25 marks, 40 minutes

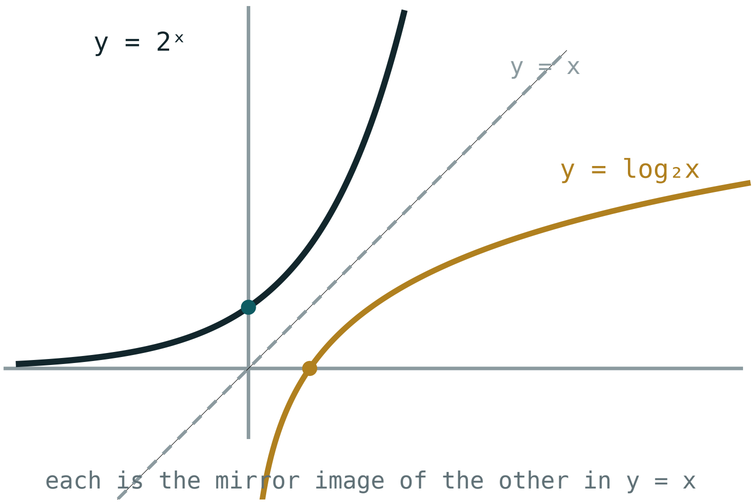
Q	TASK	MARKS	FROM
1	Evaluate $log_2 32$, $log_9 3$, $log_5 \frac{1}{25}$ and $lg\ 0.01$	4	L57–58
2	Expand $log_2 \frac{4x^3}{\sqrt{y}}$ and condense $3\ lg\ x - 2\ lg\ y$	4	L59–60
3	Solve $lg\ x + lg(x - 3) = 1$	4	L61–62
4	Solve $(log_2 x)^2 - 5\ log_2 x + 6 = 0$	4	L61–62
5	Solve $log_3(x - 1) < 2$ and $log\_(1/2)(x + 3) \geq -2$	5	L65–66
6	6 million at 11% a year: find the doubling time	4	L67–68

FOUR MARKS ARE FOR THE DOMAIN

Q3 and Q5 each give one mark for the domain line and one for the intersection or rejection. A correct final interval with no domain shown scores half.

The four errors

ERROR	LOOKS LIKE	KIND
no intersection	$log_3(x-1) < 2 \Rightarrow x < 10$, keeping $x = 0$	method
invented law	$log(x + y) = log\ x + log\ y$	knowledge
sign not reversed	$log\_(1/2)$ treated as base > 1	knowledge
root not rejected	$x = -2$ kept in $lg\ x + lg(x-3) = 1$	method



The picture that settles the direction and the domain at once.

The domain drill

Twelve expressions on the board. For each, the class calls out only **the domain**, in three seconds:

EXPRESSION	DOMAIN
$\log_2(x - 4)$	$x > 4$
$\lg(5 - x)$	$x < 5$
$\log_3(x^2)$	$x \neq 0$
$\log_5(x^2 - 4)$	$x < -2$ or $x > 2$
log_x 9	$x > 0, x \neq 1$
$\lg x + \lg(x - 3)$	$x > 3$

ROWS 5 AND 6 ARE THE ONES THEY MISS

A variable **base** has its own conditions. And two logarithms added give the **intersection** of the two domains, not the domain of the product.

02 Worked examples

EXAMPLE 1 **Model answer, Q3: solve $\lg x + \lg(x - 3) = 1$.**

1. Domain: $x > 0$ and $x > 3$, so $x > 3$.
   One mark.
2. $\lg[x(x - 3)] = \lg 10$
3. $x^2 - 3x - 10 = 0 \Rightarrow x = 5, -2$
4. $x = -2$ fails the domain.
   One mark.

ANSWER

$x = 5$

EXAMPLE 2 **Model answer, Q5b: solve $\log_{1/2}(x + 3) \geq -2$.**

1. Domain $x > -3$.
2. $-2 = \log_{1/2} 4$
3. Base < 1: reverse. $x + 3 \leq 4$.
4. $x \leq 1$; intersect.

ANSWER

$-3 < x \leq 1$

EXAMPLE 3 **Model answer, Q6: 6 million at 11% a year.**

1. $1.11^n = 2$
2. $n = \dfrac{\lg 2}{\lg 1.11}$
3. $= \dfrac{0.3010}{0.0453}$
4. ≈ 6.64 years.

ANSWER

≈ 6.6 years

03 Interactive model

Run the domain drill before the rewrite, and again at the end.

Domain, law, direction

$\log_2 32$:

- 4
- 5
- 6
- 16

$\log_9 3$:

- 0.5
- 2
- 3
- 1.5

Domain of $\lg x + \lg(x - 3)$:

- $x > 0$
- $x > 3$
- $x \neq 3$
- $\mathbb{R}$

$lg\ x + lg(x - 3) = 1$:

- $x = 5, -2$
- $x = 5$
- $x = -2$
- none

$(log_2\ x)^2 - 5log_2 x + 6 = 0$:

- $x = 2, 3$
- $x = 4, 8$
- $x = 6$
- $x = 8$

$log_3(x - 1) < 2$:

- $x < 10$
- $1 < x < 10$
- $x > 1$
- $x < 9$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Domain of definition	Aniqlanish sohasi	ОДЗ
Law of logarithms	Logarifm qoidasi	Свойство логарифма
Intersection	Kesishma	Пересечение
Careless error	E’tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo‘shlig‘i	Пробел в знаниях
Correction	Tuzatish	Исправление

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The domain line is worth:

- nothing
- a mark on its own
- the question
- half

$log(x + y)$ equals:

- $log\ x + log\ y$
- $log\ x \cdot log\ y$
- none of these
- $log\ x - log\ y$

A base under 1 in an inequality:

- changes nothing
- reverses the sign
- makes it undefined
- doubles it

Adding two logarithms gives a domain that is:

- the union
- the intersection
- the product
- wider

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $log_2 32$

   ANSWER 5

2. $log_9 3$

   ANSWER 0.5

3. $log_5 \frac{1}{25}$

   ANSWER -2

4. $lg\ 0.01$

   ANSWER -2

5. Domain of $lg(5 - x)$

   ANSWER $x < 5$

6. Condense $3\ lg\ x - 2\ lg\ y$

   ANSWER $lg \frac{x^3}{y^2}$

7. Solve $lg\ x = 2$

   ANSWER $x = 100$

Medium · the standard question

7 PROBLEMS

1. Expand $log_2 \dfrac{4x^3}{\sqrt{y}}$

ANSWER $2 + 3log_2 x - \dfrac{1}{2} log_2 y$

2. Solve $lg\ x + lg(x - 3) = 1$

ANSWER $x = 5$

3. Solve $(log_2\ x)^2 - 5log_2 x + 6 = 0$

ANSWER $x = 4, 8$

4. Solve $log_3(x - 1) < 2$

ANSWER $1 < x < 10$

5. Solve $log\_(1/2)(x + 3) \geq -2$

ANSWER $-3 < x \leq 1$

6. 6 million at 11%: doubling time

ANSWER ≈ 6.6 years

7. Domain of $log_5(x^2 - 4)$

ANSWER $x < -2$ or $x > 2$

Hard · stretch and reason

7 PROBLEMS

1. Solve $log_2(x - 1) + log_2(x + 1) = 3$

   ANSWER $x = 3$

2. Solve $lg^2 x - lg\ x^4 + 3 = 0$

   ANSWER $x = 10, 1000$

3. Solve $log\_(1/3)(x^2 - 3x) \geq -2$

   ANSWER $-1 \leq x < 0$ or $3 < x \leq 4$

4. Solve $log\_x\ 16 = 4$

   ANSWER $x = 2$

5. Solve $log_2 x + log\_x\ 2 = 2.5$

   ANSWER $x = 4$ or $x = \sqrt{2}$

6. A town of 30 000 grows 4% a year: when does it double?

   ANSWER ≈ 17.7 years

7. Explain why $log_3(x-1) < 2$ is not $x < 10$

   ANSWER The domain $x > 1$ must be intersected

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. Domain first on every question.

1. Rewrite in full every question that lost a mark, with the domain first.
2. Five problems from the section your knowledge column was heaviest in.
3. Solve $lg\ x + lg(x - 15) = 2$ and $log_2(x + 1) \leq 3$.
4. 9 million so‘m at 9% a year. Find the value after 5 years and the doubling time.